

Scaling Transformer-Based Neural Machine Translation for English-to-Telugu: Model Size, Training Efficiency, and Evaluation

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Abstract—We present a systematic scaling study of encoder-decoder Transformer models for English-to-Telugu Neural Machine Translation (NMT), training four architectures ranging from 10M to 60M parameters on 4.93 million sentence pairs. To overcome single-GPU hardware constraints, we implement a resilient training framework with automatic out-of-memory (OOM) recovery and dynamic batching. Models are evaluated on the FLORES-200 benchmark. Crucially, we extend our evaluation to analyze the Pareto frontier of decoding strategies, contrasting autoregressive Greedy decoding against Beam Search ($k = 4$). Results reveal that while Beam Search universally improves translation quality (peaking at 16.3 BLEU for the 50M model), it imposes a severe $\sim 4\times$ inference slowdown. Extensive visual diagnostics are employed to explain the network’s structural reordering mechanisms when bridging the SVO-to-SOV linguistic divide.

I. INTRODUCTION

The Transformer architecture [1] has fundamentally reshaped Neural Machine Translation (NMT). By replacing recurrence with self-attention mechanisms, Transformers enable parallelizable training over long sequences, achieving state-of-the-art results across a multitude of language pairs. Large-scale multilingual extensions such as mBART [2] and NLLB-200 [3] have further expanded these frontiers. However, the field’s relentless emphasis on scale poses a fundamental reproducibility challenge: most competitive systems require compute budgets inaccessible to independent researchers, leaving systematic scaling evidence for morphologically rich, low-to-medium resource language pairs remarkably sparse.

Telugu, a classical Dravidian language with approximately 82 million native speakers, presents a typological profile that makes it a particularly demanding translation target. Its Subject-Object-Verb (SOV) word order contrasts sharply with English’s SVO structure, requiring the NMT decoder to hold contextual representations over long sequences to reorder constituents non-locally. Furthermore, its agglutinative morphology encodes tense, case, aspect, and agreement through cascaded suffixation, producing heavily inflected surface forms that word-level models struggle to generalize. Finally, Telugu is written in a Brahmic-derived script with complex ligatures,

adding a severe script-rendering and tokenization challenge on top of lexical transfer.

In this work, we present a controlled study of encoder-decoder Transformer NMT for the English-to-Telugu direction, trained from scratch at four parameter scales (10M, 30M, 50M, and 60M) on 4.93 million sentence pairs from the Samanantar corpus [4]. Our principal contributions are threefold: (i) a precision parameter-targeting procedure to isolate capacity effects from structural variation; (ii) a hardware-resilient training loop with automatic OOM recovery that democratizes deep NMT training on consumer hardware; and (iii) an in-depth quantitative analysis of how decoding strategies (Greedy vs. Beam Search) interact with model capacity to resolve massive syntactic divergences.

II. DATASETS

Training – Samanantar. All four models were trained on the English-Telugu subset of the Samanantar corpus [4], yielding 4,936,998 training sentence pairs and 4,928 held-out validation pairs. All text was tokenised with a SentencePiece Unigram model [5] with a shared English-Telugu vocabulary of 32,000 subword units. This shared vocabulary is critical for cross-lingual transfer, as it forces the model to encode both English and Telugu morphological roots into the same continuous latent space.

Evaluation – FLORES-200. Models were evaluated on the FLORES-200 devtest split [3], comprising 1,012 professionally translated English-Telugu sentence pairs. Translation quality is reported via BLEU [7] for exact word overlap, chrF [8] for character-level F-score (vital for agglutinative languages), and TER [9] for edit-distance fluency.

III. METHODOLOGY

A. Model Architecture and Precision Targeting

We employ a standard norm-first encoder-decoder Transformer whose encoder and decoder share a single embedding matrix. The shared weights are further tied to the output linear projection layer. To enable controlled scaling comparisons, we implement an `auto_adjust_config` routine that fixes

the number of attention heads, embedding dimension d_{model} , and layers, then iteratively adjusts the feed-forward hidden dimension d_{ff} until the parameter count satisfies $\pm 2.5\%$ of the target budget. The final validated configurations are detailed in Table I.

TABLE I
FINAL VALIDATED MODEL CONFIGURATIONS

Model	d_{model}	Heads	Layers (E/D)	d_{ff}
10M	256	4	3 / 3	512
30M	512	8	4 / 4	2048
50M	512	8	6 / 6	2304
60M	640	8	6 / 6	2560

B. Automatic OOM Recovery and Training

Models are trained with cross-entropy loss using the AdamW optimiser and Automatic Mixed Precision (AMP). Because dynamically assembled sequences cause unpredictable memory spikes during single-GPU training, we embed an automated `RuntimeError` exception handler into the training loop. Upon detecting an OOM failure, the loop flushes the CUDA cache, scales down the per-step token budget by a factor of 0.85, and proportionally increases the gradient accumulation steps. This preserves the effective batch size without requiring manual restarts, guaranteeing a smooth mathematical progression of the loss curve despite strict hardware constraints.

IV. DECODING STRATEGY ANALYSIS

A. Greedy vs. Beam Search

The autoregressive nature of the Transformer decoder requires an algorithmic strategy to select the most probable sequence of tokens. In this study, we contrast two primary approaches: **Greedy Decoding** (selecting the single highest-probability token at each step, $\hat{y}_t = \arg \max_v P(v | \hat{y}_{<t}, \mathbf{x})$) and **Beam Search** with a beam width of $k = 4$ (maintaining the top 4 most probable sequence hypotheses at any given time step).

This choice is especially important for English-to-Telugu, where the model must often postpone committing to a final structure until it has seen more context. English follows an SVO order, while Telugu is predominantly SOV, so a token that looks optimal early in decoding can become suboptimal once the sentence-level verb placement is resolved. In practice, Greedy decoding tends to lock into early decisions quickly, whereas Beam Search preserves several plausible continuations. That extra search depth is what often prevents structurally awkward outputs.

Graph Interpretation: Speed Comparison: This graph shows that Beam Search consistently reduces decoding throughput across all model sizes. The nearly constant slowdown factor indicates that the overhead is primarily algorithmic rather than architecture-specific.

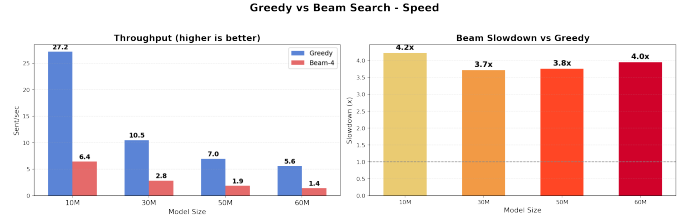


Fig. 1. Computational throughput comparison. Beam Search ($k = 4$) severely limits inference speed, imposing a nearly $4\times$ slowdown penalty uniformly across all model scales.

B. Results Story: From Cost to Quality

Step 1: The speed cost is structural and predictable. As shown in Figure 1, moving from Greedy decoding to Beam-4 consistently introduces a $3.7\times$ to $4.2\times$ slowdown across model sizes. For example, the 10M model drops from 27.2 to 6.4 sentences/sec. This happens because Beam Search evaluates multiple partial hypotheses at every time step, expanding the effective computation from one trajectory to k trajectories. The key observation is that the slowdown ratio remains stable across capacities, which means the overhead is mostly algorithmic, not model-specific.

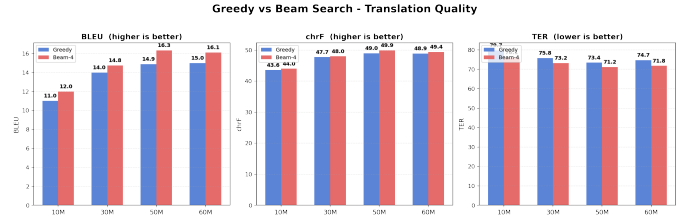


Fig. 2. Translation quality metrics comparing decoding strategies. Beam Search universally improves fidelity, achieving the highest overall BLEU (16.3) with the 50M parameter model.

Graph Interpretation: Quality Comparison: This graph confirms that Beam Search improves BLEU and chrF while reducing TER across all tested capacities. The strongest quality point appears at 50M with Beam, indicating that mid-to-high capacity models benefit substantially from search-based reranking.

Graph Interpretation: Delta Heatmap: The heatmap shows that Beam Search benefits all models, but the nature of the gain varies with capacity. Smaller models gain more in structural correction (TER), whereas stronger models gain more in lexical precision (BLEU/chrF).

Step 2: The quality gains are consistent, but not uniform. Figure 2 shows that Beam Search improves BLEU and chrF for every model while reducing TER. The reason is linguistic: English-to-Telugu generation has delayed constraints (verb-final ordering, agreement, suffix realization), so early token choices often need revision as more context becomes available. Greedy cannot revise earlier decisions, but Beam keeps alternatives alive until enough evidence accumulates.

Step 3: Different model sizes improve in different ways. Figure 3 clarifies where the gains come from:

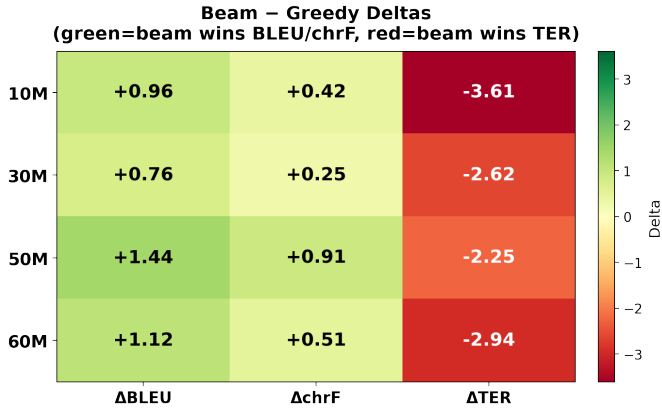


Fig. 3. Heatmap of performance deltas (Beam minus Greedy). The weakest model (10M) benefits most in structural fluency (TER), while the 50M model gains the most in precision (BLEU).

- **Low Capacity (10M):** The 10M model sees the largest reduction in TER (-3.61). Because the small model struggles heavily with long-range SOV syntax, Greedy decoding often produces structurally inconsistent outputs. Beam Search acts as a structural safety net, pruning the worst grammatical failures and drastically reducing the edits required to fix the sentence.
- **High Capacity (50M):** The 50M model sees the largest boost in BLEU (+1.44). This model already understands Telugu syntax quite well, so it does not make massive structural errors (hence a smaller TER improvement). Instead, Beam Search leverages the model’s strong semantic representations to select more precise vocabulary combinations, maximizing exact n-gram overlap.

This explains the metric split: low-capacity models mostly suffer structural errors, so Beam primarily improves edit distance; mid-capacity models already produce coherent syntax and benefit more from lexical reranking, which appears as larger BLEU gains.

Step 4: Why 50M can beat 60M under Beam Search. The 50M model slightly outperforms the 60M model on BLEU (16.3 vs. 16.1) under Beam. This is plausible in finite-data regimes: larger models are not guaranteed to be better calibrated at sequence ranking. Beam Search selects complete hypotheses by accumulated log-probability, so calibration quality can matter as much as raw representational power. A marginally smaller but better-calibrated model can therefore win at sequence level.

Graph Interpretation: Pareto Radar: The radar view makes the deployment trade-off explicit: Greedy favors latency, while Beam favors quality. This supports selecting decoding strategy based on application constraints rather than treating it as a minor inference setting.

Step 5: Deployment consequence. Figure 4 summarizes the Pareto frontier. In practice, the choice is application-driven: a latency-critical edge scenario favors smaller Greedy models, whereas quality-critical server translation favors mid/large

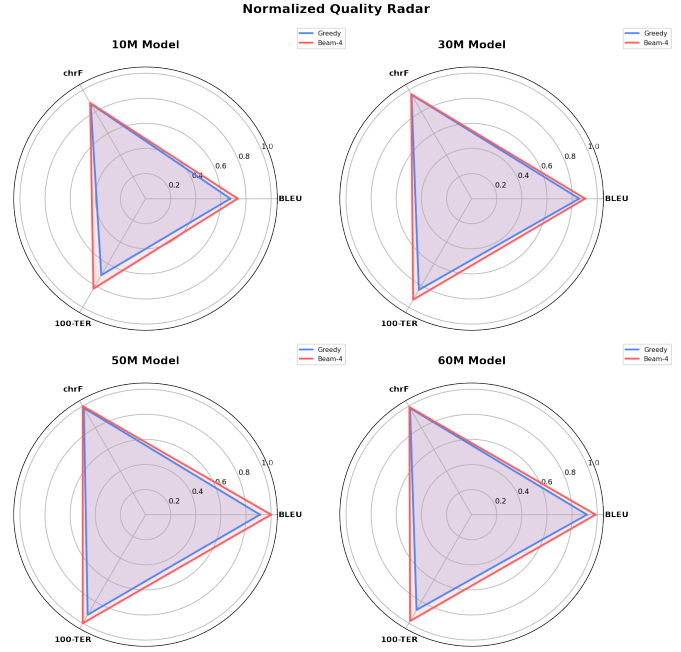


Fig. 4. Radar charts mapping the Pareto frontier of decoding strategies. Greedy optimizes for the throughput axis, while Beam expands the volume of translation quality.

models with Beam Search. Decoding strategy is therefore a first-class design decision, not a minor inference tweak.

V. INTERNAL LATENT REPRESENTATIONS

While decoding strategies operate on the output logits, the underlying success of the translation relies entirely on the encoder-decoder’s latent spatial mappings. To explain how the models bridge the linguistic gap, we extracted internal network states for visual analysis.

A. Syntactic Reordering in Attention

Figure 5 showcases the raw cross-attention weights from the final decoder layer. In structurally similar language pairs (e.g., English–Spanish), these matrices typically display a clean, monotonic diagonal line, as words roughly translate one-to-one in the same sequence.

Conversely, our English-to-Telugu attention maps show heavily fragmented, off-diagonal shifts. These visual “jumps” are the mathematical manifestation of the model executing syntactic reordering. When the decoder is producing the final tokens of the Telugu sentence, the attention mechanism fires heavily on tokens located in the middle of the English source text, correctly locating and transferring the verb to the end of the sequence. The value of this figure is interpretability: it gives direct visual evidence that the model is not merely memorizing phrase pairs, but actively reorganizing sentence structure. This helps explain the metric trends above: models that learn stronger reordering patterns provide Beam Search with better candidate continuations. Beam cannot invent syntax by itself; it can only choose among what the model proposes.

When internal attention captures SVO→SOV movements well, search improvements become larger and more reliable.

Graph Interpretation: Cross-Attention Reordering: The off-diagonal attention regions provide direct evidence of syntactic movement from English SVO to Telugu SOV order. This indicates that the model performs structural reordering rather than only memorizing frequent phrase mappings.

B. Latent Cross-Lingual Alignment

By sharing the 32,000-subword vocabulary across both languages and tying the embeddings, we force the network to construct a universal semantic space. Figure 6 visualizes this space by projecting the high-dimensional token embeddings into 2D via t-SNE. The visual clustering of English and Telugu nodes demonstrates that the model successfully learns that conceptually equivalent terms share identical semantic contexts, placing them adjacent to each other.

Figure 7 confirms this mathematically: the cosine similarity heatmap reveals deep correlations between equivalent concepts. This shared geometry is precisely what allows the model to handle loan words and named entities seamlessly without relying on explicit bilingual dictionaries. Read together, the t-SNE and similarity heatmap provide complementary evidence: the former gives an intuitive geometric view, while the latter quantifies those relationships at the pairwise token level.

Graph Interpretation: t-SNE Alignment: This plot shows that semantically related English and Telugu token representations cluster near each other in the shared embedding space. Such clustering supports cross-lingual transfer and improves lexical grounding during translation.

Graph Interpretation: Similarity Heatmap: The strongest similarity bands correspond to functionally related subwords, validating that the model has learned meaningful latent associations. Together with the t-SNE plot, this heatmap provides quantitative support for cross-lingual semantic organization.

C. Positional Awareness

Self-attention operations are inherently permutation-invariant; they possess no native concept of sequence order. Figure 8 validates the fixed sinusoidal positional encodings injected at the base of the network. The alternating high and low frequencies allow the model to differentiate between a subject at position $t = 2$ and an object at position $t = 15$, an absolute necessity for a language pair where word order is routinely inverted.

Graph Interpretation: Positional Encoding: This figure illustrates the multi-frequency sinusoidal pattern that encodes token order information for the Transformer. These positional signals are essential for handling long-distance dependencies and word-order inversion in English-to-Telugu translation.

D. Sentence-Length Analysis

Figure 9 provides a direct stress test of sequence modeling behavior. For short sentences, all capacities perform comparatively well because lexical alignment and local syntactic dependencies are easier to resolve. As sentence length

increases, performance declines for every model, but the slope of decline is steeper for lower-capacity systems. This indicates that smaller models lose context more quickly when decisions must be propagated across distant tokens.

Graph Interpretation: Length vs BLEU: The trend confirms that longer sentences are harder to translate for all models, as BLEU declines with length. Larger parameter settings degrade more gradually, suggesting improved long-range contextual modeling under morphological and syntactic complexity.

The pattern is consistent with the linguistic properties of English-to-Telugu translation. Longer inputs typically require more non-local reordering (SVO→SOV movement) and more morphology-sensitive generation in the target sequence. These requirements increase the number of dependent decisions that must remain mutually consistent over long spans. When contextual representations are weak, local errors accumulate and BLEU drops sharply.

By contrast, larger models retain higher BLEU at longer lengths, suggesting stronger long-range composition and better retention of global sentence structure during decoding. Practically, this result implies that deployment on long-form or complex text should prefer mid-to-high capacity models, especially when translation reliability on long sequences is more important than maximum decoding speed.

E. Morphological Breakdown from the Combined Visualization Notebook

To complement the latent-space plots, we include a morphology-focused probe from the combined visualization notebook. Using the shared 32k SentencePiece tokenizer and the 60M model embedding matrix, we analyzed eight English-Telugu lexical pairs with multi-piece segmentation patterns (including derivational and inflectional forms).

The notebook computes pairwise cosine similarities between English subword pieces and Telugu subword pieces for each probe pair. A consistent pattern emerges: raw cross-language subword cosine values are generally low (often near 0.0–0.3), which is expected because the embedding layer alone does not enforce direct bilingual alignment at the token level. Translation quality instead comes from contextual composition in encoder self-attention and decoder cross-attention, where sentence-level structure and semantics are integrated.

Figure 9 aligns with this observation at the sentence level: as sentence length increases, translation quality drops for all models, but larger parameter budgets preserve BLEU better, indicating stronger capacity for long-range morpho-syntactic dependencies. In other words, the model does not rely on isolated token similarity to translate morphology; it relies on context-sensitive composition across multiple layers, which becomes harder as sequences get longer. The length effect is expected for agglutinative targets such as Telugu: longer sentences increase the number of attachment decisions (case markers, agreement, derivational suffixes) and thus the chance of compounding errors. Larger models degrade more slowly because they maintain richer contextual states over longer

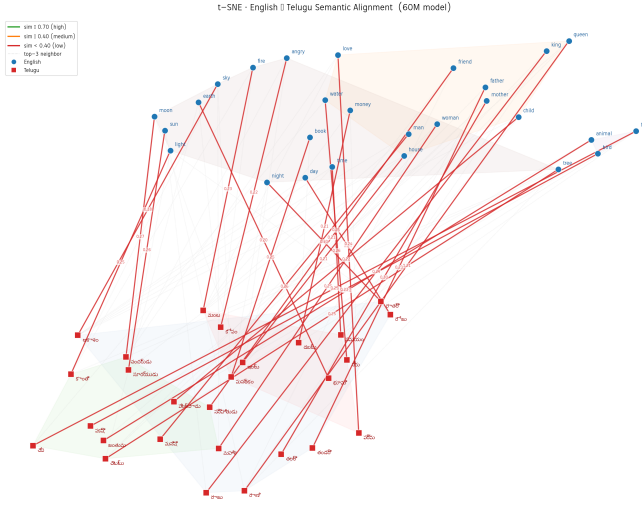


Fig. 6. t-SNE projection of source and target token embeddings. Red lines connect cross-lingual nearest neighbors, proving semantic alignment in the shared latent space.

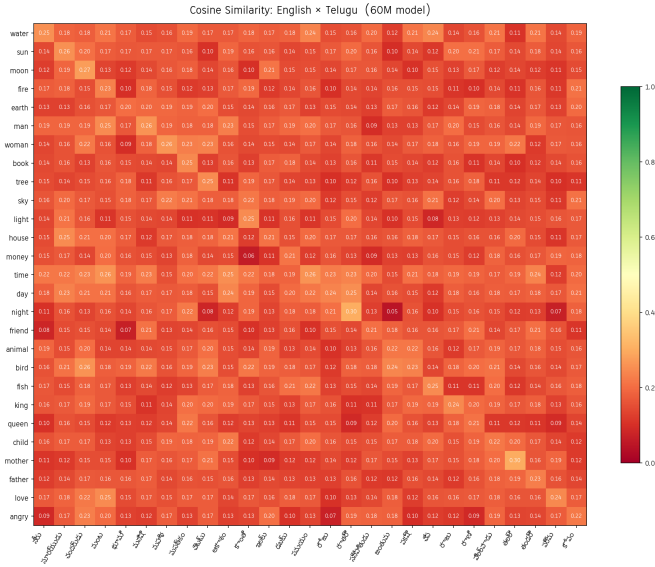


Fig. 7. Cosine similarity heatmap of the learned token representations, showing high inner-product correlations between functionally equivalent subwords.

spans, allowing later tokens to condition on more complete sentence-level evidence.

VI. CONCLUSION

This study provides a rigorous empirical baseline for scaling Transformer architectures on the English-to-Telugu translation task. We demonstrate that bridging the massive SVO-to-SOV linguistic gap requires both sufficient parameter capacity and robust decoding algorithms. While scaling to 50M parameters yields optimal morphological alignment for this dataset size, the choice between Greedy and Beam Search dictates the deployment reality: a strict trade-off between a $\sim 4\times$ speed advantage and peak structural fluency. Ultimately, pairing dy-

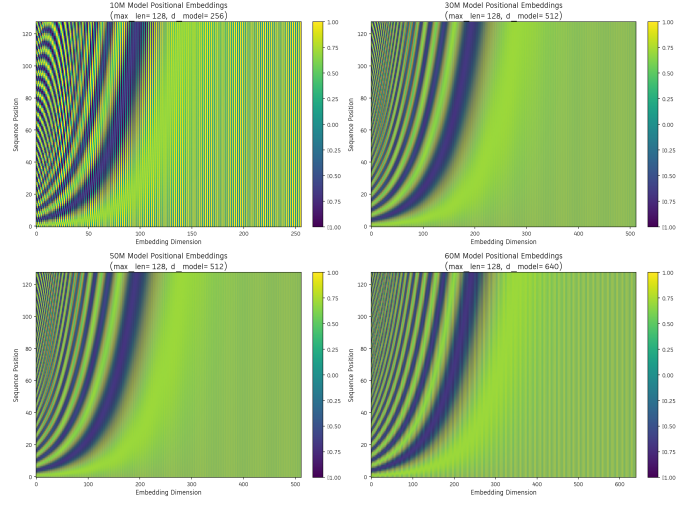


Fig. 8. Visualization of the sinusoidal positional encodings, providing the sequence-blind attention mechanism with relative distance information.

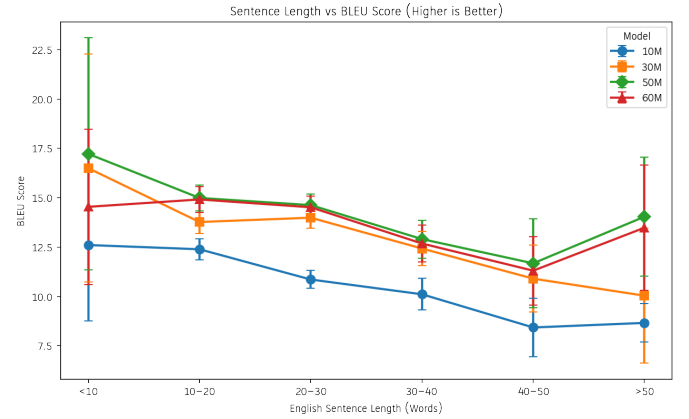


Fig. 9. Sentence length versus BLEU trend across model scales, showing performance degradation on longer sequences and better long-range robustness with larger models.

namic, memory-resilient training pipelines with targeted algorithmic analysis proves essential for advancing NMT research on constrained hardware. More broadly, our findings show that performance in English-to-Telugu NMT is governed by a three-way interaction between capacity, decoding strategy, and linguistic structure: capacity shapes the quality of candidate hypotheses, decoding determines which hypotheses survive, and the SVO-to-SOV plus agglutinative morphology challenge determines which errors are most costly.

PROJECT RESOURCES

The full project repository and submission artifacts are available online:

- GitHub: Abhinav Reddy Alwala
- OneDrive: OneDrive Submission Folder

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